

DevOps Lab Program

Program-5: Demonstrate CI integration by running Jenkins in a Docker container

Program Structure:

Program-5/

→Dockerfile

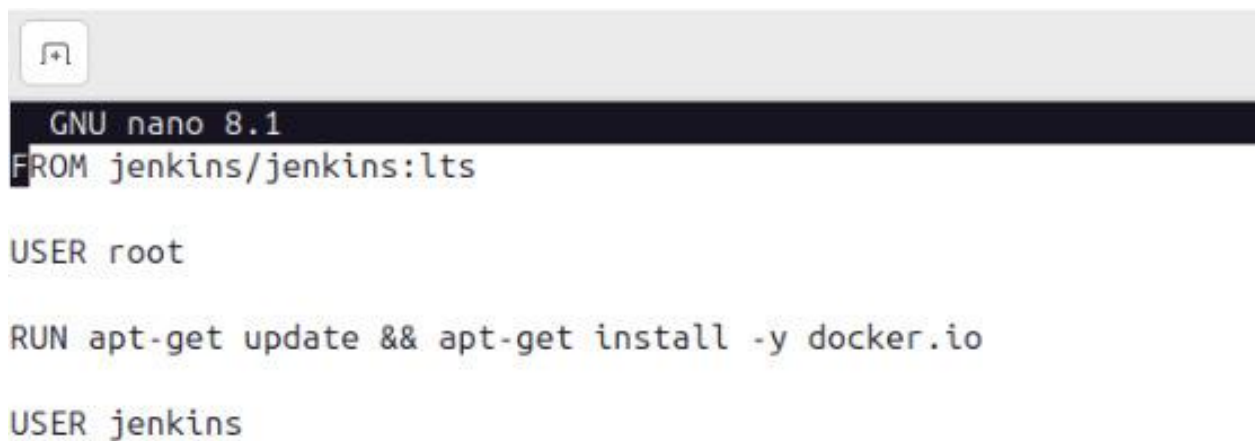
→docker-compose.yaml

Steps for Execution:

Step 1: Check whether docker service is running

Step 2: Create a folder program-5 and navigate to it

Step 3: Create a Dockerfile



```
GNU nano 8.1
FROM jenkins/jenkins:lts

USER root

RUN apt-get update && apt-get install -y docker.io

USER jenkins
```

Step 4: Create a docker-compose.yaml with following contents

```
GNU nano 8.1
version: "3.8"

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 5: Build and run the jenkins

```
Nov 19 16:14
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program5

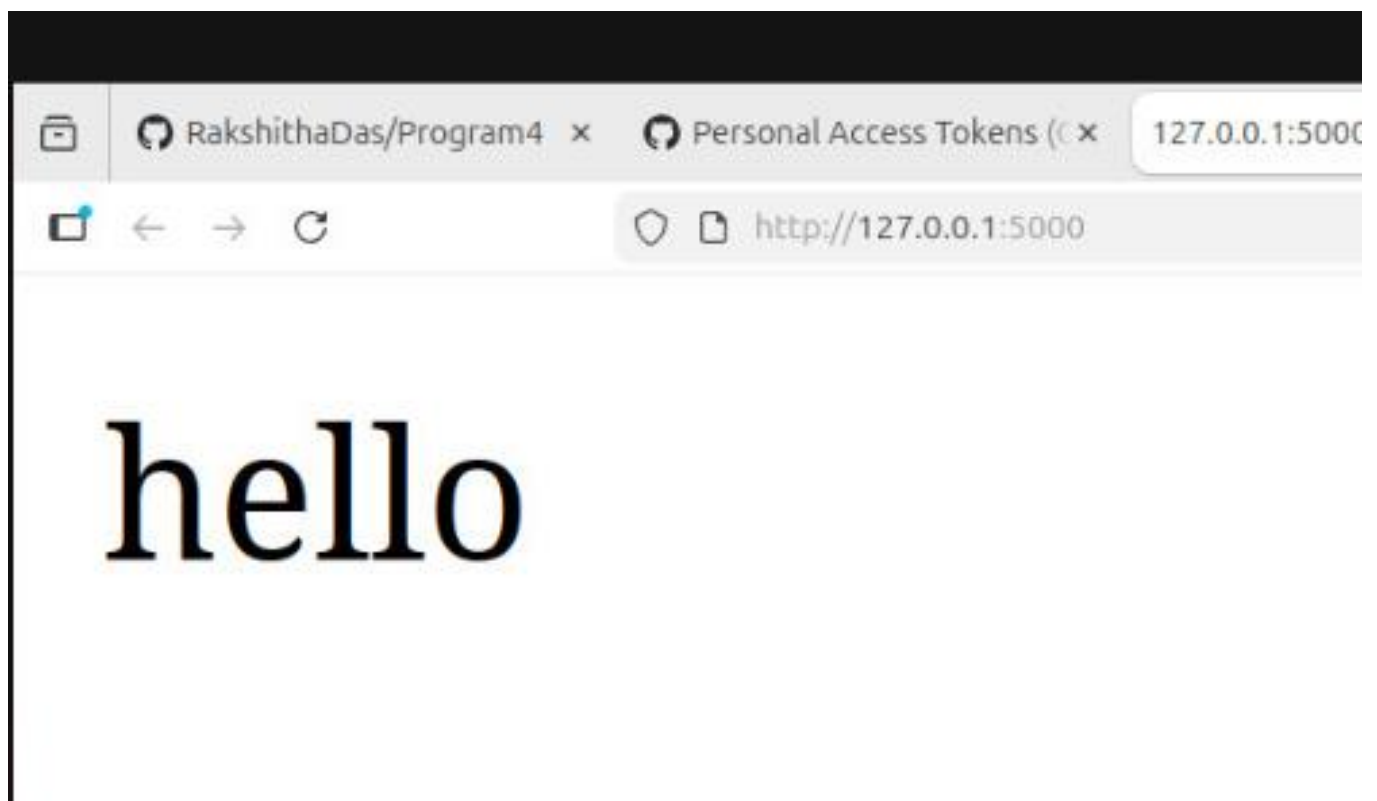
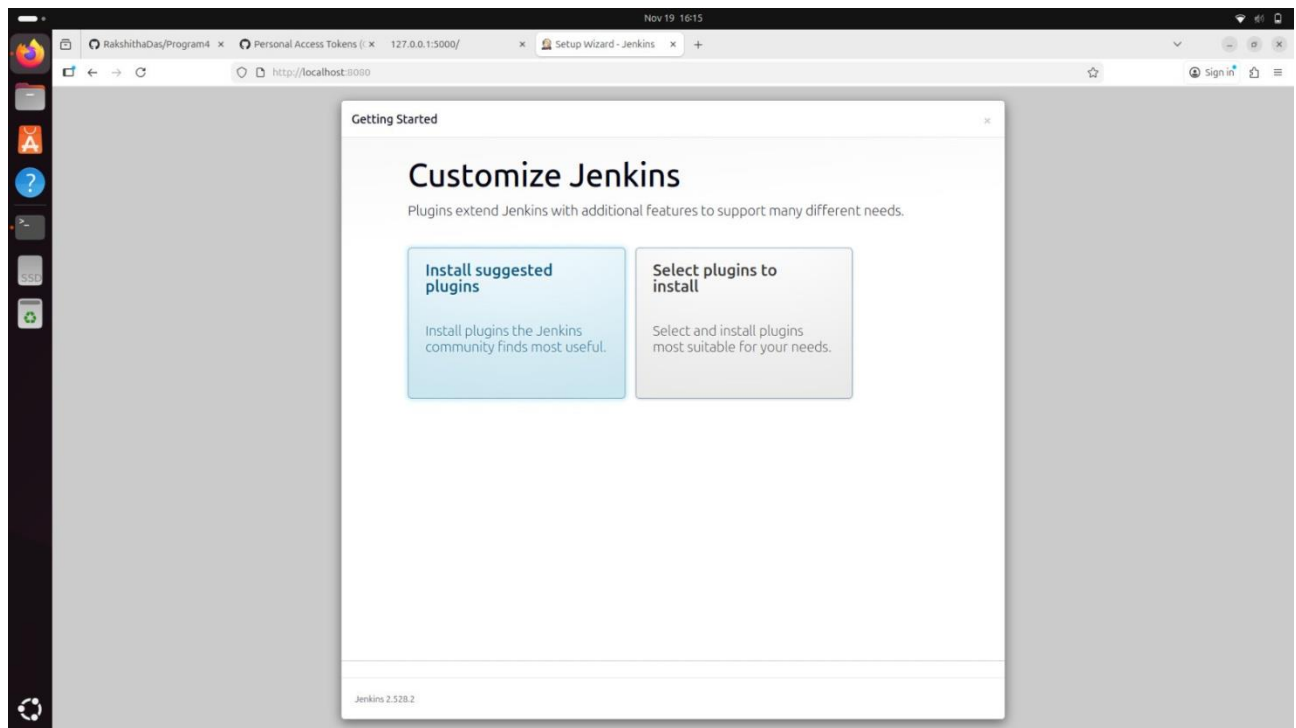
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program4
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program3
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program5

2025-11-19 10:34:20.153+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
2025-11-19 10:34:20.256+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oejs9n.ContextHandler$CoreContextHandler@3c782d8e[Jenkins v2.528.2,/,b=file:///var/jenki
ns_home/war/,a=AVAILABLE,h=oejs9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@572e6fd9(STARTED)]
2025-11-19 10:34:20.267+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@7bca6fac(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2025-11-19 10:34:20.282+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@3668d4[STARTING][12.0.25,sto=0] @2222ms
2025-11-19 10:34:20.283+0000 [id=40] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2025-11-19 10:34:20.422+0000 [id=39] INFO jenkins.model.Jenkins#<init>: Starting version 2.528.2
2025-11-19 10:34:20.503+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2025-11-19 10:34:20.514+0000 [id=55] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2025-11-19 10:34:21.194+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2025-11-19 10:34:21.199+0000 [id=55] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2025-11-19 10:34:21.200+0000 [id=57] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2025-11-19 10:34:21.318+0000 [id=47] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2025-11-19 10:34:21.319+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2025-11-19 10:34:21.319+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2025-11-19 10:34:21.320+0000 [id=50] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2025-11-19 10:34:21.346+0000 [id=75] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-11-19 10:34:21.707+0000 [id=51] INFO jenkins.install.SetupWizard#init:

[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> d4be112e13c84a19a986b62a5d18f471
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****

2025-11-19 10:34:40.073+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2025-11-19 10:34:40.098+0000 [id=39] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2025-11-19 10:34:44.502+0000 [id=75] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2025-11-19 10:34:44.503+0000 [id=75] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
2025-11-19 10:35:34.748+0000 [id=41] INFO winstone.Logger#logInternal: JVM is terminating. Shutting down Jetty
2025-11-19 10:35:34.749+0000 [id=41] INFO org.eclipse.jetty.server.Server#doStop: Stopped oejs.Server@3668d4[STOPPING][12.0.25,sto=0]
2025-11-19 10:35:34.760+0000 [id=41] INFO o.e.j.server.AbstractConnector#doStop: Stopped ServerConnector@7bca6fac(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2025-11-19 10:35:34.760+0000 [id=41] INFO hudson.lifecycle.Lifecycle#onStatusUpdate: Stopping Jenkins
2025-11-19 10:35:34.772+0000 [id=41] INFO jenkins.model.Jenkins$1#onAttained: Started termination
```

Step 6: Verify whether jenkins is running by navigating to <http://localhost:8080>



Step 7: Display the docker images and remove the program5 image